

AURORA-ADAS Perception Module — Whitepaper Edition

1. Overview

The Perception subsystem transforms raw sensor data into actionable signals for navigation, safety, and world-model prediction. It includes:

- **DroNet** — monocular steering + collision probability
- **Depth Estimator** — dense geometry
- **Semantic Segmentation** — class-aware scene understanding
- **TensorRT Runtime** — FP16/INT8 optimized inference
- **Training + Evaluation Pipelines**
- **Benchmarking + Deployment Tools**

This module is optimized for **Jetson Orin / Xavier / Nano** and integrates tightly with:

- Mapping
- Planning
- Safety
- World-Model

2. DroNet Architecture

DroNet is a lightweight CNN designed for real-time UAV navigation.

Inputs:

- 200×200 RGB image

Outputs:

- Steering $\in [-1, 1]$
- Collision probability $\in [0, 1]$

Backbone:

- ResNet-8 style convolutional encoder
- Dual-head regression network

Deployment:

- FP16 TensorRT engine (default)
- INT8 TensorRT engine (ultra-fast)

3. Perception Pipeline

1. RGB → DroNet
2. DroNet → Steering + Collision
3. RGB → Depth Estimator → Depth Map
4. RGB → Semantic Segmentation → Mask
5. RGB → ViT Encoder → Latent Embedding
6. Latent → Dynamics → z_{t+1}
7. z_{t+1} → Free-Space / Risk / Steering Enhancer

Outputs feed:

- Local Mapper
- Global Mapper
- MPC Local Planner
- Safety Supervisor
- World-Model Node

4. Training Pipeline

Includes:

- AirSim dataset generator
- Dataset builder
- Training script
- Evaluation suite
- INT8 calibration dataset builder

5. Benchmark Summary

Precision	Latency (ms)	FPS	Accuracy Drop
FP32	6.8	147	baseline
FP16	2.1	476	+0.5%
INT8	1.1	910	+2.1%

6. Jetson Deployment

1. Install JetPack
2. Copy models
3. Launch DroNet
4. Switch FP16/INT8 engines
5. Optimize clocks + power mode

7. RViz Visualization

Includes:

- RGB
- Depth
- Semantic mask
- Steering plot
- Collision plot
- Local map
- Global path

8. Ablation Study

- Backbone depth
- Steering-only vs dual-head
- Quantization effects
- Collision head contribution

9. Ethical Considerations

DroNet is intended for **safe autonomous navigation**, not surveillance or harmful applications.

10. Maintainer

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